MIL-L-29400B(MC)
26 July 1984
SUPERSEDING
MIL-L-29400A(MC)
18 November 1983

MILITARY SPECIFICATION

LAUNCHER, MINE CLEARANCE: MARK 155

This specification is approved for use by U.S. Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification establishes the requirements for manufacture and Government acceptance of the Mark 155 Mine Clearance Launcher.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-704	- Treatment and Painting of Materiel
MIL-I-6866	- Inspection, Penetrant Method of
MIL-I-6868	- Inspection Process, Magnetic Particle

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commandant of the Marine Corps (LMA-1), Headquarters, U.S. Marine Corps, Washington, D.C. 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-454	- Standard General Requirements for Electronic Equipment
MIL-STD-781	- Reliability Tests Exponential Distribution
MIL-STD-794	- Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	- Environmental Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. MARINE CORPS

82A5052A0000	- Launcher Assembly for M58 Linear Demolition Charge, Trailer Mounted

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951 - Commercial Packaging, Practice For

(Application for copies of ASTM Publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), sample units shall be subjected to first article inspection (see 4.4 and 6.3). The first article shall be manufactured using the same methods, procedures, and tooling proposed for production. Approval of first article by the procuring activity shall not relieve the contractor of the obligation to supply equipment conforming to this specification. Changes or deviations in production units from the approved first article baseline shall require written authorization by the procuring activity.

3.2 Materials. Materials used in fabrication of the launcher shall be as specified on Data List (DL) 82A5052A0000. Materials shall be free of defects and imperfections that might affect the serviceability and appearance of the finished product. Substitute material shall not be utilized prior to written authorization by the procuring activity (see 4.6.1).

3.3 Design and construction. The launcher shall be fabricated, assembled, and adjusted in accordance with DL 82A5052A0000 (see 4.6.2).

3.3.1 Welding. The launcher and its components shall be welded in accordance with DL 82A5052A0000 (see 4.6.15).

3.3.2 Launch rail/rocket interface. The rocket rail shall interface with the MK 22 MOD 3 rocket without any physical interference during loading or firing (see 4.6.8).

3.4 Maintainability. The maximum time to repair shall be no greater than 0.5 hours at the organizational level and 2.0 hours at the direct or general support level, which is defined as the 90th percentile of the repair times at the appropriate maintenance levels (see 4.6.17).

3.5 Performance characteristics. The launcher shall meet or exceed the following performance requirements.

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3.5.1 Operational profile. The launcher shall be capable of a 10-mile cross-country mission with one rocket launch and one line charge detonation at the conclusion of the 10-mile trip. The 10 miles shall be apportioned as follows:

(a) 70 percent gravel and dirt roads -- at 35 miles per hour (mph) maximum..

(b) 20 percent level cross-country --- at 15 mph maximum.

(c) 10 percent hilly cross-country --- at 10 mph maximum.

The reliability of the launcher for the mission described shall be at least 0.94 (94 per cent) (see 4.6.16).

3.5.2 Launch rail positioning. The launch rail shall be capable of attaining the following positions:

(a) Container loading ---- 75° ± 5°.

(b) Launching ---------- 47° ± 2°.

(c) Stowed ---------- 0° to 5° (no pressure in hydraulic cylinder).

All angular measurements of the launch rail shall be with respect to the horizontal axis of the main launch support (see 4.6.7 and 4.6.10).

3.5.3 Launch rail raising (hand pump). The launch rail shall be capable of being vertically positioned by use of the hydraulic hand pump anywhere between the stowed position and the container loading position, as specified in 3.5.2, without binding or distortion (see 4.6.7).

3.5.4 Launch rail raising (accumulator). With the accumulator pressurized to 2900 pounds per square inch gage (psig) maximum and when remotely actuated by pulling the lanyard attached to the hydraulic valve handle, the launch rail (loaded with the Government-loaned inert rocket) shall raise to the launch position specified in 3.5.2 within 4 seconds. The loaded launch rail shall operate smoothly without binding or distortion (see 4.6.10).

3.5.5 Hydraulic fluid loss. In the static condition, the hydraulic system, pump seals, valves, fittings, hoses, etc., shall show no evidence of fluid beyond the seal points other than seal weep or seep. Seal weep or seep is the evidence of fluid beyond the seal which does not result in the formation of a droplet. During the operational mode (pressurizing accumulator, raising

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and lowering launch rail), 1 drop per operation shall not be exceeded at operational devices such as valves, cylinder and pump. Hydraulic lines, hoses, and fittings shall show no evidence of leakage other than seal weep or seep as defined for the static system (see 4.6.11).

3.5.6 Hydraulic valve. The hydraulic valve shall perform the following functions:

(a) Manual raise/lower - With the hydraulic valve positioned to connect the hand pump to the cylinder, the launch rail shall be capable of being raised and lowered, as specified in 3.5.3, when actuated by the hand pump and the integral release valve respectively (see 4.6.7).

(b) Pressurize accumulator - With the hydraulic valve positioned to connect the hand pump to the accumulator, the accumulator shall be capable of being pressurized by actuation of the hand pump. The hydraulic valve shall entrap hydraulic fluid in the accumulator at 3000 ± 100 psig for a minimum of 2 hours without a loss of more than 25 psig (see 4.6.9).

(c) Remote raise - with the hydraulic valve positioned to connect the accumulator to the cylinder, the launch rail shall be capable of being raised by pulling the lanyard which rotates the hydraulic valve from pressurize accumulator position to remote raise and allows the accumulator to discharge into the cylinder (see 4.6.10),

3.5.7 Ball-lock safety pin. The ball-lock safety pins shall be capable of being inserted and removed in the launch support "LOCK" position holes to prevent movement of the launch rail up or down. The ball-lock safety pins shall also be capable of being inserted in the "RAISE" position holes to limit upward travel of the launch rail to the launch position specified in 3.5.2 (see 4.6.12).

3.5.8 Safety switch. The safety switch shall complete the firing circuit to the MK 22 MOD 3 rocket when the launch rail is at an elevation of 40° ± 2°. The firing circuit shall remain completed for elevations above 38° (see 4.6.13).

3.5.9 Rocket restraint device. The rocket restraint device shall release the MK 22 MOD 3 rocket when the launch rail is at an elevation of 35° ± 2°. The restraint device shall remain released for elevations above 33° (see 4.6.14).

3.5.10 Environmental requirements. The launcher shall withstand the following operating and storage environmental conditions (See 4.6.18).

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3.5.10.1 Operating. The launcher shall perform all of the requirements specified in 3.5.2 through 3.5.9 while exposed to the following environmental conditions:

(a) Temperature: -25F (-32°C) to + 125°F (52°C) in accordance with MIL-STD-810, Method 501.1, Procedure II and Method 502.1, Procedure I.

(b) Vibration: Level and duration as specified in MIL-STD-810, Method 514.2, Procedure XIII, Part 2.

(c) Shock: Level and duration as specified in MIL-STD-810, Method 516.2, Procedure I, shock parameters A and C of Figure 516.2-2.

(d) Dust: Level and duration as specified in MIL-STD-810, Method 510.1, Procedure I.

(e) Salt Fog: Level and duration as specified MIL-STD-810, Method 509.1, Procedure I.

(f) Rain: Level and duration as specified in MIL-STD-810, Method 506.1, Procedure II.

(g) Freezing rain: Top and sides of launcher with accumulation of 0.25 ± 0.06 inch of ice at 20°F (-7°C).

(h) Solar radiation: Level and duration as specified in MIL-STD-810, Method 505.1, Procedure I.

3.5.10.2 Storage. The launcher shall perform all of the requirements specified in 3.5.2 through 3.5.9 following exposure to the following environmental conditions:

(a) Temperature: -65°F (-54°C) to +140°F (60°C) in accordance with MIL-STD-810, Method 501.1, Procedure I and Method 502.1, Procedure I.

(b) Vibration: Level and duration as specified in 3.5.10.1.b, above.

(c) Shock: Level and duration as specified in 3.5.10.1.c, above.

3.6 Painting. The launcher and its components shall be painted in accordance with DL 82A5052A0000 and MIL-T-704, type F or type G, as applicable. The paint shall be applied in accordance with MIL-T-704. The paint thickness shall be in accordance with MIL-T-704. Surfaces after painting shall meet the requirements of MIL-T-704 (see 4.6.3).

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3.7 Marking/serialization.

3.7.1 Launcher. The launcher shall be marked and serialized as specified on DL 82A5052A0000 (see 4.6.4).

3.7.2 Parts and components. Parts and components shall be marked in accordance with MIL-STD-130 (see 4.6.4).

3.7.3 Identification plates. The launcher identification plates shall be in accordance with DL 82A5052A0000 (see 4.6.5).

3.8 Government-furnished property. When specified (see 6.2), the following property shall be furnished by the Government.

Item No.	Item	Quantity
1	Blasting Machine, M34 NSN 1375-00-567-0223	1
2	Blasting Cap Tester, M51 NSN 4925-00-999-3454	1
3	Trailer Chasis, M353, MS53031-1 NSN 2330-00-542-2831	1

3.9 Government-loaned property. The following property shall be loaned by the Government for first article conformance tests:

Item No.	Item	Quantity
1	Linear Demolition Charge, INERT, M68A2	1
2	Rocket, MK 22 MOD 3 (certified inert by the Government and ballasted to identical weight of a live rocket)	1

3.10 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9 (see 4.6.6).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract of purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements

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specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).

(b) Quality performance inspection (see 4.5)

4.3 Inspection conditions. All inspections shall be performed in accordance with the requirements specified in 3.5.10.1.

4.4 First article inspection. First article inspection shall be performed following the award of contract and prior to production, at a location acceptable to the Government. Inspection shall be performed on the sample units which have been produced in accordance with 3.1. Failure of the first articles to comply with 4.4.1 due to manufacturing defects or procedures shall be cause for refusal of acceptance by the Government, until measures satisfactory to the Government have been taken and a successful retest has been performed.

4.4.1 First article examinations and tests. All first article units shall be subjected to the examinations and tests specified in 4.6. No defects shall be allowed.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the examinations and tests specified in 4.6.

4.5.1 Lot size. An inspection lot shall consist of all launchers producted during an identifiable production period and submitted at one time for acceptance inspection. Production period shall be 1 month or as specified in the contract or purchase order (see 6.2).

4.5.2 Sampling for inspection. Sampling for inspection from each inspection lot shall be in accordance with inspection level N-II-S of MIL-STD-105. The acceptable quality level (AQL) shall be 0.65 percent for critical defects, 1.0 percent for major defects and 2.5 percent for minor defects. Classification of defects is indicated in 4.6.

4.6 Examinations and tests.

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4.6.1 Material certification. (201) The contractor shall certify that all materials and parts used in fabrication of the launcher are in accordance with the requirements of 3.2. Material or part manufacturer's certification of test data will be acceptable for fulfilling this requirement. Acceptance or approval of any material or part shall not be construed as a guarantee of acceptance of the launcher. The contractor shall be responsible for using parts and assemblies from Qualified Products Lists (QPL's) whenever available. Contractor's inspection records shall specifically list all QPL items by number and date of the QPL, name of contractor, and part number or drawing number(s). When parts and assemblies are approved as qualified products, but not yet listed on the QPL, the contractor shall list the products by number and date of the approved document, and name of supplier(s).

4.6.2 Construction. (202) The launcher shall be visually examined to verify proper assembly and installation of components and assemblies in accordance with the requirements of 3.3.

4.6.3 Painting. (203) The launcher shall be inspected to verify conformance to the requirements of 3.6.

4.6.4 Marking/serialization. (204) The launcher shall be visually examined to the requirements of 3.7.1 and 3.7.2.

4.6.5 Identification plates. (205) The launcher identification plates shall be visually examined to verify conformance to the requirements of 3.7.3.

4.6.6 Workmanship. (206) The launcher shall be visually examined to verify conformance to the requirements of 3.10.

4.6.7 Launch rail raising (hand pump). (101) The launch rail shall be elevated from stowed position to the container loading position and returned to the stowed position to determine conformance to the requirements of 3.5.2, 3.5.3, and 3.5.6.a. This test shall be performed five times on each launcher tested.

4.6.8 Launch rail/rocket interface. (1) The launch rail/rocket interface requirements of 3.3.2 shall be verified by use of inspection gage, Drawing No. 82A5052A9100. Any physical or misalignment shall be considered a defect.

4.6.9 Hydraulic valve. (102) Using the hydraulic hand pump, the accumulator shall be pressurized to determine conformance to the requirements of 3.5.6.b.

4.6.10 Launch rail raising (accumulator). (103) The loaded rocket launch rail shall be elevated from the stowed position to the launching position and returned to determine conformance to the requirements of 3.5.2, 3.5.4 and 3.5.6.c. This test shall be performed five times on each launcher tested.

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4.6.11 Hydraulic fluid loss. (104) Visual examination of the hydraulic system shall be conducted after each cycle of 4.6.7, 4.6.9, and 4.6.10 to determine conformance to the requirements of 3.5.5. Seal weep or seep is acceptable.

4.6.12 Ball-lock safety pin. (105) To determine conformance to the requirements of 3.5.7, visual and mechanical examination shall ascertain proper operation of the ball-lock safety pin in addition to the required usage in conjunction with 4.6.10.

4.6.13 Safety switch. (2) An electrical continuity check of the electrical circuit shall be performed before and after each raising of the rocket launch rail. The angle of the launch rail at which the safety switch actuates shall be in accordance with the requirements of 3.5.8.

4.6.14 Rocket restraint device. (3) Visual examination of the rocket restraint device shall be performed before and after each raising of the rocket launch rail to ascertain proper performance. The angle of the launch rail at which the rocket restraint device releases the rocket shall be in accordance with the requirements of 3.5.9.

4.6.15 Welding inspection. (106) Structural welds shall be inspected in accordance with MIL-I-6866 or MIL-I-6868 to determine conformance to the requirements of 3.3.1.

4.6.16 Operational test. (107) Unless otherwise specified in the contract or purchase order (see 6.2), tests performed to demonstrate conformance to operational requirements of 3.5.1 shall be the responsibility of the Government. First article reliability testing shall be conducted in accordance with MIL-STD-781, test plan VC, with a minimum acceptable mean - cycles - between failure (0_1) of 3.83. Testing shall be conducted to the mission profile of 3.5.1.

4.6.17 Maintainability verification. (108) Unless otherwise specified in the contract or purchase order (see 6.2), tests performed to demonstrate conformance to maintainability requirements of 3.4 shall be the responsibility of the Government.

4.6.18 Environmental tests. (109) Unless otherwise specified in the contract or purchase order (see 6.2), tests performed to demonstrate conformance to environmental requirements of 3.5.10 shall be the responsibility of the Government.

4.6.18.1 Freezing rain. The launcher with inert rocket installed, shall be subject to freezing rain test. The freezing rain shall be produced in a cold chamber by soaking the test item at 0°F(-17.8°C) for 12 hours. The chamber temperature shall then

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be increased to 20°F(-6.7°C) and the test item sprayed with water precooled to 35°F(1.7°C) at a rate of 1 inch (25.4mm) per hour to cover the top and sides with 0.25 ± 0.06 inch of glaze. The launcher shall be cycled to determine conformance to 3.5.10.1.g.

4.6.19 Inspection of packaging. The preservation, packaging, packing, and container marking shall be inspected, prior to shipment, to determine conformance to Section 5 of this specification.

5. PACKAGING

5.1 Preparation. Prior to packaging, all launchers shall be drained of hydraulic fluid.

5.2 Preservation and packaging. Preservation and packaging shall be level B, C, or industrial, as specified (see 6.2).

5.2.1 Level B. The complete launcher shall be preserved and packaged in accordance with the level B requirements of MIL-STD-794.

5.2.2 Level C. The complete launcher shall be preserved and packaged in accordance with the level C requirements of MIL-STD-794.

5.2.3 Industrial. The complete launcher shall be preserved and packaged in accordance with ASTM-D-3951.

5.3 Packing. Packing shall be level B, C, or industrial (see 6.2).

5.3.1 Level B. The complete launcher, preserved and packaged as specified in 5.2.1, shall be packed to conform to the level B requirements of MIL-STD-794.

5.3.2 Level C. The complete launcher, preserved and packaged as specified in 5.2.2, shall be packed to conform to the level C requirements of MIL-STD-794.

5.3 Industrial. The complete launcher packaged as specified in 5.1.2 shall be packed in conformance with ASTM-D-3951.

5.4 Marking. Marking for packaging and packing shall be in accordance with MIL-STD-129 or ASTM-D-3951, as specified (see 6.2).

6. NOTES

6.1 Intended use. The launcher described herein will enable rapid breaching of minefields and other obstructions implaced on inland areas.

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6.2 Ordering data

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.

(b) First article, if required (see 3.1).

(c) Operational test, if required (see 4.6.16).

(d) Maintainability verification, if required (see 4.6.17).

(e) Environmental tests, if required (see 4.6.18).

(f) Level of preservation and packaging (see 5.2 and 5.3).

(g) Type and amount of Government furnished property (see 3.8).

(h) Length of production period determining lot size (see 4.5.1).

(i) Marking (see 5.4).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified shall be delivered as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (N) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph No.	Data Requirement Title	Applicable DID No.	Option
3.1	First Article Inspection Procedure	DI-T-4901	-
3.1	First Article Inspection Report	DI-T-4902	-
4.6	Test Plan	DI-T-4903	-
4.6	Test Report	DI-T-4904	-

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(Data item descriptions related to this specification, and identified in Section 6 will be approved and listed as such in DoD 5000.19L, VOL. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specified acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of two units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article (see 3.1).

6.4 Government-furnished property. The contracting officer should arrange to furnish the property listed in 3.8 as specified in 6.2.

6.5 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.9.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:

Navy - MC

Project No. 1055-N073

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-29400B(MC)	LAUNCHER, MINE CLEARANCE: MARK 155

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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